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2. Each receptor may activate several adenylate cyclases each of which make much c AMP.
3. Thus there is more signal after each step.

The actions of cAMP are terminated by phosphodiesterases. The effect of peptide hormones like insulin, glucagon, somatotropin are usually short lived because they work through second messenger system.

- Steroid hormones can easily cross the cell membrane, and bind to their receptors, which are intracellular or intranuclear. Upon binding to the receptors, they pair up with another receptor – hormone complex (dimerize).This dimer can then bind to DNA and alter its transcription. (Figure 11.18)

Avoid use of steroid components

The abuse of anabolic steroids can cause serious health problems like high BP, heart diseases, liver damage, cancer, stroke and blood clots. Other side effects of steroid use includes nausea, vomiting, ligament and tendon injuries, head ache, joint pain, muscle cramps, diarrhoea, sleep problem etc.

Basal metabolic rate (BMR): The amount of energy needed to keep the body at rest.

- The effect of steroid hormones such as aldosterone, oestrogen, FSH are long lived, as they alter the amount of mRNA and protein in a cell.
- Amino acid derived hormones are derived from one or two aminoacid with a few additional modifications. Thyroid hormone is synthesised from tyrosine and includes the addition of several iodine atoms. Epinephrine an amino acid derivative may function through second messenger system like peptide hormones or they may actually enter the cell and function like steroid hormones.

Activity

1. Prepare a chart to show the location of various endocrine glands of human.
2. Visit a nearby medical college or hospital and gather data of endocrine disorder, diseases and treatment measures.

On an educational tour to Nilgiris, Nitish and his friends observed the local people. Few of them were with swollen neck.

1. Is it a disease? If so what is the cause? How is the disease treated?

Visit a nearby medical college or hospital and gather data of endocrine disorder, diseases and treatment measures.

When my Aunt comes to India from America through air after a long journey, she suffers with jetlag. Find out the reason.